Year 12 Maths - Statistical hypothesis testing

Similar questions - One-tailed binomial hypothesis tests

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A sweet manufacturer claims that 15% of its sweets are red. Mia thinks the proportion is actually higher. She takes a random sample of 40 sweets and finds that 11 are red.

(a) State suitable hypotheses and carry out a test at the 5% significance level. **[6]**

(b) Find the critical region for this test and state the actual significance level. **[3]**

(c) Mia's friend counts 10 red sweets in a different random sample of 40. State the conclusion of the test for this sample. **[1]**

2 Historically, 60% of patients recover within a week from an illness. A new drug is given to 25 randomly chosen patients. The doctor wants to test, at the 5% significance level, whether the drug increases the recovery rate.

(a) State suitable hypotheses and find the critical region for the test. **[4]**

(b) State the probability of a Type I error for this test. **[1]**

(c) In the trial, 19 patients recover within a week. State the conclusion of the test in context. **[2]**

(d) Explain why a conclusion of "accept H_0" does not prove that the drug has no effect. **[1]**

Question 1

(a) $H_0 : p = 0.15$, $H_1 : p > 0.15$, where p is the proportion of red sweets. Under H_0, $X \sim B(40, 0.15)$. $P(X \geqslant 11) = 1 - P(X \leqslant 10) = 0.0299$. Since $0.0299 < 0.05$, reject H_0: there is evidence that the proportion of red sweets is greater than 15%

(b) $P(X \geqslant 10) = 0.0672 > 0.05$ and $P(X \geqslant 11) = 0.0299 < 0.05$, so the critical region is $X \geqslant 11$. Actual significance level 2.99%

(c) 10 is not in the critical region (or $0.0672 > 0.05$), so do not reject H_0: insufficient evidence that the proportion is higher than 15%

Question 2

(a) $H_0 : p = 0.6$, $H_1 : p > 0.6$. $X \sim B(25, 0.6)$. $P(X \geqslant 19) = 0.0736 > 0.05$ and $P(X \geqslant 20) = 0.0294 < 0.05$. Critical region $X \geqslant 20$

(b) $P(\text{Type I error}) = P(X \geqslant 20 \mid p = 0.6) = 0.0294$

(c) 19 is not in the critical region, so do not reject H_0: there is not sufficient evidence at the 5% level that the drug increases the recovery rate

(d) Not rejecting H_0 only means there is insufficient evidence; the sample could be too small to detect a real improvement